

Anil Pradhan

Backend Engineer | AI Systems Builder

Computer Science undergraduate passionate about building scalable backend systems and AI-powered applications. Experienced in FastAPI, LangGraph, RAG pipelines, and cloud deployment. Eager to contribute to impactful products and grow as a software engineer.

anilpradhan9644@gmail.com
+91 8917476908
Bhubaneswar, Odisha, India
linkedin.com/in/anil-pradhan543
github.com/Anil-Pradhan-web
my-portfolio-anil.vercel.app

Education

ITER, SOA University, Bhubaneswar
B.Tech in Computer Science & Engineering
2024 - 2028 | CGPA: 9.11/10

Odisha Board of Secondary Education
Class XII (PCM with IT)
2022 - 2024 | Percentage: 85%

Experience

College Ambassador | Techfest, IIT Bombay
Jun 2026 - Present

- Selected as a Campus Ambassador for Techfest, IIT Bombay, participating in technical and industry-oriented initiatives.
- Contributed to web development and marketing by successfully completing multiple project-based assignments.
- Conducted market research, prepared a business strategy presentation, and qualified for Round 2 of the FlowFinance Growth Strategy & Business Development Challenge.

Technical Projects

AI Career Mentor | Live Demo | Solo Full-Stack Developer
March 2026 - Present

- Multi-Agent Orchestration & Concurrency:** Orchestrated 7 specialized AI agents using a parallel LangGraph DAG; reduced end-to-end latency by 60% (from 4 mins to 60s) by executing resume parsing and market scraping concurrently.
- System Resilience & Fault Tolerance:** Built failover architecture with custom Circuit Breakers and LLM fallback chains (Groq ⇌ Cerebras ⇌ NVIDIA NIM), preventing request drops during API rate-limit spikes.
- Real-time Bidirectional Streaming & State Machines:** Designed a full-duplex WebSocket protocol with Gemini Multimodal Live API for low-latency audio coaching; built a 7-phase FSM coding interview simulator with Monaco IDE.
- High-Efficiency RAG & Semantic Search:** Optimized ChromaDB RAG using local ONNX embeddings (all-MiniLM-L6-v2) to eliminate API costs; built a keyword-search fallback to prevent OOM container crashes.
- Rate-Limiting, Caching & Telemetry:** Integrated Upstash Redis middleware for sliding-window rate limiting and JWT validation; built a telemetry dashboard tracking API latencies and token cost rollups.
- Data Durability, Dockerization & CI/CD:** Enforced data validation with Pydantic v2, persisting entries to Postgres with PgBouncer pooling; deployed multi-stage Docker builds with a 116 Pytest CI/CD suite.

FastAPI LangGraph ChromaDB ONNX Docker Upstash Redis Next.js 14 Gemini Live

Task Pilot-AI | Live Demo | Integration Lead • Team IdeaForg-E
June 2026 - July 2026

- Solved Developer Context Overload:** Aggregated engineering tasks from 7 sources (Jira, GitHub, Slack, Email, Calendar, Meetings, Incidents) into a unified DB, saving developers ~2 hours daily.
- Recovered Untracked "Hidden" Tasks:** Developed an LLM and rule-based extraction system to parse emails and Slack chats for hidden action items, capturing ~35% of previously untracked tasks.
- Reduced LLM Costs and API Failures:** Built local fallback algorithms to reduce token costs and bypass Groq API rate limits; used fuzzy text-matching for deduplication and local Slack parsing.
- Prevented Database Locking and Workflow Freezes:** Configured SQLite in WAL mode to allow concurrent multi-agent writes without locking; built a self-healing monitor to auto-restart stuck pipeline runs.
- Automated Real-Time Emergency Re-Prioritization:** Created a chat command API allowing developers to inject critical P1 incidents; autonomously re-ran the 6-stage pipeline to update schedules in under 15s.

FastAPI LangGraph SQLite LLMs React Redis

Technical Skills

Languages:	Python, Java, JavaScript, SQL	Frontend:	HTML, CSS, Tailwind CSS, Axios, React
Backend:	FastAPI, LangGraph, Pydantic, SQLAlchemy 2.0, WebSocket, Prometheus, Sentry, pdfplumber, edge-tts	AI / LLM & RAG:	Groq Cloud, Cerebras Cloud, NVIDIA NIM, Gemini Live API, Tavily, Serper.dev, DuckDuckGo
Databases:	Neon Postgres, SQLite, Upstash Redis, ChromaDB (Vector Store)	Auth & Security:	JWT (python-jose), Google OAuth 2.0, bcrypt, CORS Middleware, Rate Limiting, HTTPS
DevOps & Infra:	Docker, GitHub Actions (CI/CD), Vercel, Render.com	Tools:	VS Code, Git, GitHub, Swagger UI, Postman

Achievements & Certifications

- Google Cloud Arcade Program (2024-25): Completed GCP challenges, earning 75 Arcade Points across hands-on cloud labs.
- Solved 200+ problems across GFG and LeetCode, awarded GFG Certificate and T-shirt.
- Certifications: Credentials in Python, AI, and Web Dev from IBM, Cisco, Anthropic, Infosys, Udemy.
- Competitions: Participated in multiple IIT/NIT coding contests and hackathons focused on AI systems, backend engineering, and algorithmic problem solving.
- Hackathons: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+), Amazon Nova AI (\$95K+), and Google Solution Challenge 2026.
- Active participant in Developer Communities (GDG Events) and college tech events, exploring AI systems, cloud infrastructure, and backend engineering.